MOHAMED SLIMANI

Final-Year Mechanical Engineering Student | Seeking a 6-Month PFE Internship (Jan/Feb 2027)

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Professional Profile

Final-year Mechanical Engineering student with hands-on experience in industrial valve design, automotive connector development, finite element analysis (FEA), CNC manufacturing and Lean process improvement. Skilled in parametric CAD, technical drawings, DFM/DFMA and mechanical prototyping, with Python and Excel/VBA skills for engineering calculations and data analysis. Seeking a six-month PFE internship in mechanical design, simulation, manufacturing or process improvement, starting January/February 2027.

Education

Engineering Degree in Mechanical Engineering (in progress) 2024–2027 | Graduation expected 2027

Integrated Preparatory Programme – Mathematics and Physics 2022–2024

Professional Experience

Mechanical Designer – Industrial Valve Systems Toronto, Canada
Guide Valve Limited Feb 2026–01 Dec 2026

- Design industrial ball-valve assemblies comprising 150+ components using CATIA V5 and SolidWorks; prepare detailed manufacturing drawings and bills of materials (BOMs).
- Develop parametric CAD models and use Excel/VBA macros for engineering configurations, file organization and STEP/drawing exports; apply dimensional and functional tolerancing.

Product Engineering Intern – MQS Connector Redesign Tangier, Morocco
TE Connectivity 01 Jul 2026–01 Sep 2026

- Redesigned an automotive MQS connector's external locking feature through functional analysis, concept evaluation, mechanical calculations and parametric CAD.
- Evaluated locking and snap-fit concepts using ANSYS Mechanical FEA, considering insertion/retention forces, dimensional constraints, injection-molding feasibility and DFM/DFMA.
- Developed Excel/VBA calculation sheets and supported prototype and mechanical validation planning.

STEM Product Designer Fes, Morocco
MOBDIE – EdTech & Robotics Mar 2026–Jun 2026

- Designed and prototyped educational mechanical systems, including a walking robot and water-wheel mechanism, focusing on mechanical integration and ease of assembly.
- Prepared components for FDM 3D printing and laser cutting using Fusion 360 and DXF manufacturing files.

Process Optimization Engineering Intern Nouaceur, Casablanca
Flyson – Agricultural Drone Services Aug 2025–Sep 2025

- Reduced drone mission preparation time by 60% through process analysis and Lean methods (SMED, 5S and Kaizen).
- Built Python data-analysis tools and standardized DJI Agras T50 field operations through SOPs, KPIs and Odoo ERP workflows.

Mechanical Design & Manufacturing Intern Fes, Morocco
Advanced Manufacturing Laboratory Feb 2025–Jul 2025

- Designed and manufactured an ASTM B213 powder-flowability test bench from concept to functional prototype.
- Carried out CAD/CAM and fabrication tasks using CATIA V5, Fusion 360 CAM, Haas VF-2 CNC machining and FDM/SLS 3D printing.

Technical Skills

Mechanical Design: CATIA V5, SolidWorks, Creo, Fusion 360, parametric CAD, engineering drawings, GD&T, BOMs, functional analysis, DFM/DFMA.
Simulation & Analysis: ANSYS Mechanical, FEA, static/modal analysis, ANSYS Fluent, CFD, MATLAB/Simulink, heat transfer, fluid mechanics.
Manufacturing: CAD/CAM, CNC machining (Haas VF-2), Fusion 360 CAM, G-code, FDM/SLS additive manufacturing, laser cutting, prototyping.
Process Improvement & Programming: Lean Manufacturing, SMED, 5S, Kaizen, SOPs, KPIs, Python, Excel/VBA macros, Power BI, Odoo ERP.
Languages: Arabic (native), English (C1), French (B2), Spanish (A2).

Selected Engineering Projects

- **Aerospace Fuselage Structural Optimization – EMAN Aerospace**: Led a six-student team on structural design, engineering calculations and static/modal FEA; placed 2nd among external participating teams.
- **Patient-Specific Cranial Implant & Mold Design**: Reconstructed cranial geometry from 3D scans; designed implant interfaces and a two-part mold for prototype manufacturing.
- **Tube Bending Machine**: Developed functional analysis, machine architecture, structural sizing and detailed design of the bending mechanism and frame.

Leadership & Activities

Member Experience Manager – AIESEC Fes 2022–2025 (2.5 years)
Managed recruitment, performance evaluation and HR coordination activities in an international student organization of 100+ members.

Vice President – Student Office (BDE) 2025–2026
Coordinated events, student initiatives and budget management within the engineering school.